
Part II

Heating Process Optimization

Chapter 10

Background

This chapter provides the necessary background to introduce terms and concepts that may be unfamiliar to the reader, thereby facilitating the understanding of the work presented in this second part of the thesis.

10.1 Physics Informed Machine Learning

Informed Machine Learning (IML) refers to a learning paradigm in which learning is guided by a combination of data and prior knowledge. In this paradigm, the prior knowledge is provided by an independent source and explicitly incorporated into the machine learning pipeline [71].

Physics-Informed Machine Learning (PIML) is a specific instance of this paradigm, in which machine learning models are trained to perform learning tasks while incorporating knowledge of the physical laws governing the phenomenon under study.

Prior physical knowledge can be incorporated into a neural network at different stages of the learning process, including the training data, the network architecture or model structure, and the learning algorithm [16]. These different integration strategies result in different classes of physics-aware neural networks:

- *Physics-Guided Neural Networks* rely on training datasets generated from experiments or numerical simulations that already satisfy the underlying physical laws. In this case, physical knowledge is incorporated implicitly through the data used for training;

- *Physics-Encoded Neural Networks* incorporate physical knowledge directly into the network architecture or model structure by designing its components and connections according to known physical laws;
- *Physics-Informed Neural Networks* incorporate physical knowledge into the learning process through the loss function, where governing equations, boundary conditions, or conservation laws are included as additional terms.

10.2 Forward problem and inverse problem

In the scientific field, the difference between a forward problem and its inverse lies in the direction of the inference. In the former, the objective is to predict the distribution of observations given known distributions of the initial conditions and model parameters. In the latter, the objective is to infer the distributions of the initial conditions and model parameters from the available observations. Consequently, the inverse problem can be ill-posed when the available observations are insufficient to determine a unique solution.

As illustrated in Figure 10.1, in physics, the forward problem can be solved using numerical simulators since the physical laws governing the phenomenon under study are known. When, on the other hand, the inverse problem is considered, this information is not available, and only the observations can be used. Consequently, numerical simulators cannot be directly used to solve the inverse problem, and alternative strategies, such as neural networks, can be adopted.

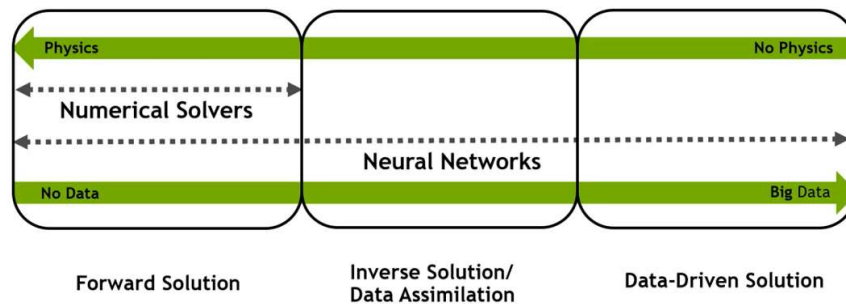


Figure 10.1: Solving forward problem and inverse problem

Chapter 11

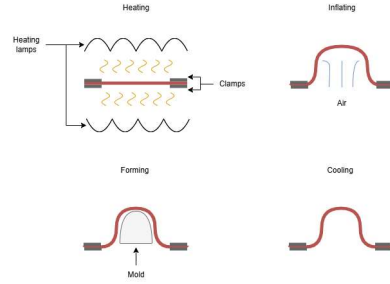
Introduction

Thermoforming is a process in which a plastic sheet is heated and shaped over a mold, transforming it from a rigid to a malleable state [64]. In industrial applications, this process is automated using a thermoforming machine and typically involves four stages: (1) heating, (2) inflating, (3) forming, and (4) cooling, as shown in Figure 11.1a. The heating phase is critical for the final product quality: if the sheet temperature is too high, material degradation or sagging may occur, whereas if it is too low, the material may be insufficiently deformable [55, 52].

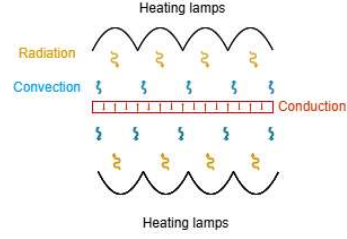
The final geometry obtained after cooling is strongly influenced by the thermal conditions established during heating, particularly the temperature distribution at the core of the sheet. However, direct measurement of the temperature distribution at the core of the sheet is not feasible in industrial settings, where only surface measurements are typically available.

As shown in Figure 11.1b accurate modeling of the heating phase requires accounting for heat conduction within the sheet, convection with the surrounding environment, and radiative exchange between lamps and the sheet. Since radiative heat transfer depends on the fourth power of temperature and on geometric view factors [7], high-fidelity numerical models must solve coupled heat transfer equations to predict the temperature distribution within the oven [66].

Although physics-based simulators that reproduce both machine control logic and heat transfer phenomena can accurately estimate the internal temperature distribution, their computational cost prevents their use in real-time control loops.



(a) The four stages of thermoforming

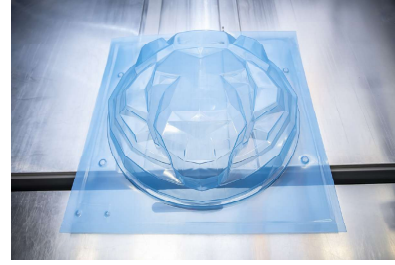


(b) Thermoforming heating phase: conduction, convection, and radiation

Figure 11.1: Overview of thermoforming: process stages and heat transfer mechanisms during the heating phase



(a) Thermoforming machine



(b) Thermoformed product result

Figure 11.2: Thermoforming machine and an example of a thermoformed product obtained at the end of the process

Given a specific mold geometry, such as the one shown in Figure 11.2b, domain experts aim to apply greater heating to protruding regions of the sheet, which require increased flexibility, while limiting heating in other areas. Experts can therefore qualitatively define a desired core temperature distribution; however, achieving it typically requires multiple trial-and-error iterations to determine suitable machine parameters, resulting in significant time and material waste.

To address these challenges, we propose two artificial neural network (ANN) models. The first is a surrogate model of a physics-based simulator, designed to provide fast and accurate predictions of the temperature distribution at the core of the sheet. This surrogate model is then used to train a second network, an inverse model that predicts machine parameters from a desired temperature distribution while explicitly accounting for process cost.

Although the inverse model can reduce process costs, its predictions may deviate from the target temperature distribution due to the ill-posed nature of the inverse problem, in which different machine parameter configurations may produce similar temperature distributions. Reducing these deviations is important to improve the robustness of the model, which is trained on simulator-generated data and must be applied to real-world cases that may exhibit significant variability in product geometry and forming requirements.

To this end, a targeted optimization step is applied at the level of individual instances. This procedure refines the inverse model solution by reducing the temperature distribution error while adjusting the trade-off between thermal accuracy and process cost.

One of the main challenges in applying data-driven approaches in industrial settings is data scarcity. As noted by Wanigasekara et al. [76], thermoforming can be considered a small-data problem due to the high cost and time required for data acquisition. To address this limitation, we generated a dataset using a simulation tool provided by SCM Group S.p.A., which combines a physics-based simulator with a coupled machine control model. A total of 18 432 samples were generated using *Latin Hypercube Sampling* (LHS) [38], of which 16 384 were used for training and 2 048 for validation. The test set consists of 45 real industrial cases collected by the industrial partner during one day of testing.

This study was conducted using an industrial thermoforming machine as reference, shown in Figure 11.2a, equipped with 240 halogen lamps arranged in two grids of 120 elements each, positioned above and below the sheet.

The remainder of this second part of the thesis is organized as follows. Chapter 12 presents the state of the art; Chapter 13 presents the surrogate model; Chapter 14 describes the inverse model; Chapter 15 introduces the optimization strategy; and Chapter 16 concludes this part of the thesis.

Chapter 12

State of the art

The use of neural networks for solving physical problems has been extensively studied in the literature, particularly following the introduction of Physics-Informed Neural Networks (PINNs) by Raissi et al. [50].

A significant portion of PINN approaches and data-driven methods in thermoforming has focused on inverse modeling strategies aimed at identifying optimal process parameters to reduce defects in the final product geometry. For instance, Yang et al. [80] proposed an inverse neural network model to predict optimal process conditions for achieving a uniform thickness distribution across the sheet. Chang et al. [12] developed an inverse model that maps desired product dimensions to process parameters in order to minimize dimensional deviations. Similarly, Leite et al. [29] trained a neural network using processing parameters as inputs and defect indicators as outputs, incorporating an optimization objective to reduce forming defects. Tan et al. [63] employed dual neural networks to infer process parameters from simulation images and optimize them for shear angle uniformity.

More recent studies have shifted attention toward earlier stages of the thermoforming process, particularly the heating phase. Simoncini et al. [59] proposed a neural network model to predict polymer behavior under coupled thermal and mechanical effects, with emphasis on infrared heating interactions. Römer et al. [51] applied reinforcement learning to design a temperature control strategy for the pre-forming stage in automated tape laying processes. Jalilvand et al. [22] used deep reinforcement learning to optimize heating settings for a target temperature

distribution, achieving reduced energy consumption. Ayçiçek et al. [2] developed an artificial intelligence-based thermal inspection system using thermal imaging, reducing defects and material waste. Turan et al. [66] proposed a surrogate model based on Convolutional Neural Networks (CNNs) and addressed the inverse problem of predicting lamp powers using gradient-based optimization.

Despite these advances, existing approaches either rely on surface temperature information or do not explicitly model the temperature distribution at the core of the sheet, which is critical for product quality. Moreover, these methods often assume simplified industrial conditions, such as uniform sheet properties or single-sided heating configurations, and frequently rely on steady-state assumptions that fail to capture the transient dynamics of the heating phase, limiting their applicability in real industrial environments.

In this study, we address these limitations by explicitly modeling the temperature distribution at the core of the sheet under realistic industrial thermoforming conditions. In contrast to prior work, the proposed method does not rely on steady-state assumptions; instead, it is trained on time-dependent simulations to accurately predict the final temperature distribution.

Chapter 13

Surrogate model

The proposed surrogate model $\mathcal{S}$ is designed to approximate the physics-based simulator by learning the mapping from machine parameters and sheet properties to the resulting core temperature distribution. Its objective is to provide accurate and computationally efficient predictions that can be integrated into the heating optimization pipeline.

To capture the spatial information of the thermoforming process, $\mathcal{S}$ is implemented using a *Mesh Graph Neural Network* (MGNN) [46], a class of neural networks capable of producing predictions that account for the mesh structure of the input elements.

MGNNs operate on a graph representation $G = (V, E)$, where nodes V correspond to mesh vertices and edges E encode local geometric relationships. Each node is associated with a feature vector x_i , while each edge feature e_{ij} encodes relative spatial information such as displacement and distance.

The MGNN architecture consists of three stages: encoder, processor, and decoder (Figure 13.1). During the encoding stage, both node features x_i and edge features e_{ij} are mapped into their respective embedded representations, v_i^E and e_{ij}^E . The processor stage performs message passing [3] over the graph, where information is exchanged between nodes and their neighbors via the connecting edges over m time steps, leading to iterative updates of the node embeddings v_i^E . Finally, in the decoding stage, the updated node embeddings are mapped to the model's outputs, denoted by v'_i , which represent the network's prediction at mesh vertex i .

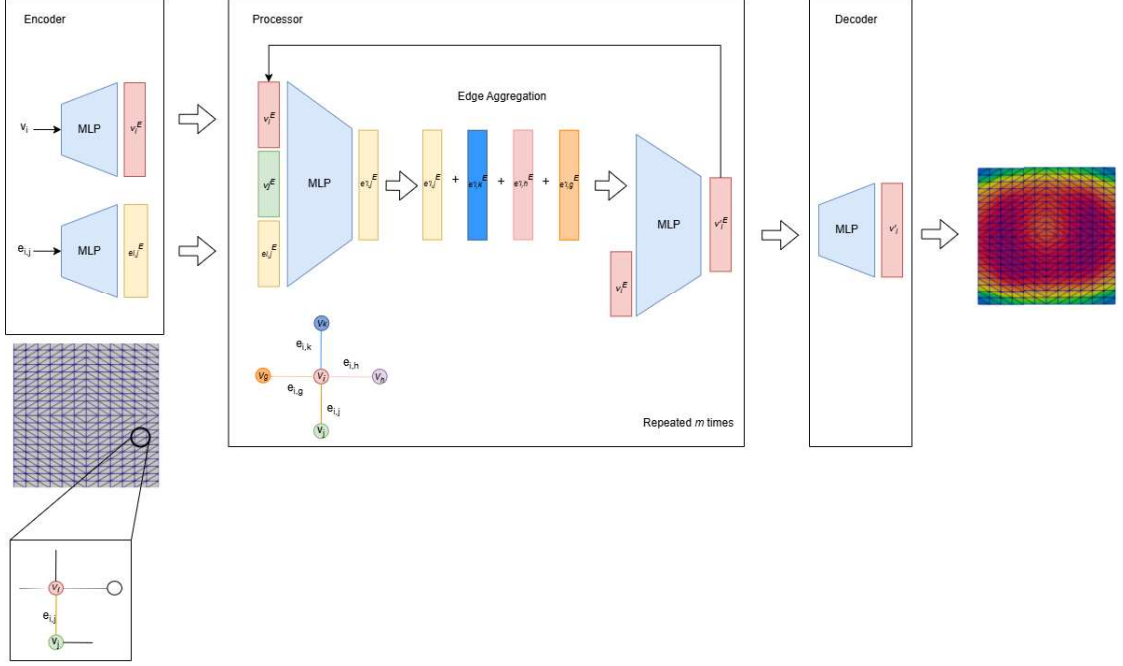


Figure 13.1: Mesh Graph Neural Network (MGNN) representing the surrogate model architecture.

The input vector of $\mathcal{S}$ consists of 254 features describing mesh node coordinates, sheet geometry (width, height, and thickness), material properties (volumetric heat capacity, emissivity, and thermal conductivity), pyrometer setpoints, lamp powers, and modulation parameters controlling the heating cycle.

The model outputs the predicted core temperature distribution over the mesh vertices together with four process-related quantities: energy consumption, heating time, and the maximum temperatures on the top and bottom surfaces of the sheet.

The encoder consists of two fully connected layers with 512 units, while the decoder consists of two fully connected layers with 128 units. ReLU activations are used throughout. The model is trained using the loss function $\mathcal{L}_s$ defined in Eq. 13.4, optimized with the Adam optimizer [25] using a learning rate of 1×10^{-3} and a batch size of 64. Training is initially set to 1000 epochs, with early stopping applied based on the validation loss, resulting in convergence after 864 epochs.

The training was performed on a workstation equipped with an Intel Core i7 CPU, 16 GB of RAM, and an NVIDIA RTX A4000 GPU, requiring approximately four days.

13.1 Loss function formulation

The loss function adopted for the training of $\mathcal{S}$ balances temperature prediction accuracy and consistency of the process-related outputs. Let $\hat{\mathbf{y}} \in \mathbb{R}^{N \times d}$ denote the predicted outputs and $\mathbf{y}$ the corresponding ground truth, where N is the number of mesh vertices and d is the output dimension. The first component of the output corresponds to the temperature distribution $\hat{\mathbf{y}}_C$, while the remaining $d - 1$ components correspond to the process-related outputs. The temperature distribution error is defined as the relative ℓ_2 norm:

$$E_T = \frac{\|\hat{\mathbf{y}}_C - \mathbf{y}_C\|_2}{\|\mathbf{y}_C\|_2} \quad (13.1)$$

while the process-related outputs error is computed through the mean squared error:

$$E_P = \sum_{k=2}^d \frac{1}{n} \sum_{i=1}^n (\hat{\mathbf{y}}_{i,k} - \mathbf{y}_{i,k})^2 \quad (13.2)$$

A sigmoid weighting term is used to balance the contributions of E_T and E_P in the final loss:

$$\sigma = \text{sigmoid} \left(\frac{E_T - \delta}{\tau \cdot \delta} \right) \quad (13.3)$$

where $\delta = 0.003$ defines the acceptable relative ℓ_2 error threshold and $\tau = 0.1$ controls the transition sharpness between the two loss regimes, both selected empirically based on preliminary experiments.

The final loss is defined as:

$$\mathcal{L}_s = \sigma \cdot E_T + (1 - \sigma) \cdot \frac{1}{E_P + 1} \quad (13.4)$$

This loss formulation adaptively prioritizes temperature distribution accuracy when errors are large, while maintaining consistency with process-related outputs once the temperature prediction is sufficiently accurate.

13.2 Model assessment

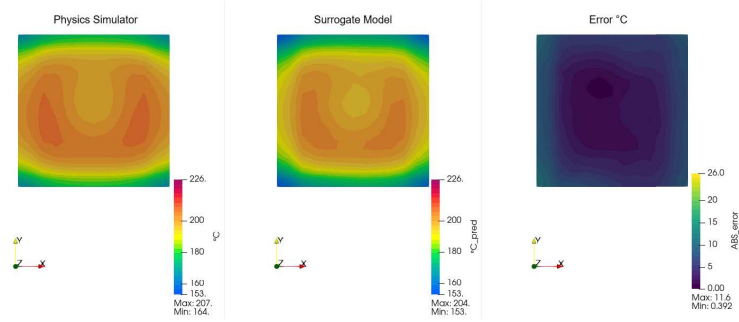


Figure 13.2: Prediction example on the test set. From left to right: the temperature distribution obtained from the physics-based simulator, the prediction produced by the surrogate model and the absolute error between the target temperature distribution and the predicted one.

At the end of training, evaluation of the temperature distribution reveals a maximum average absolute error of 1.23 °C on the training set, 3.43 °C on the validation set, and 9.65 °C on the test set.

Figure 13.2 shows an example prediction from the test set. In the example, the surrogate model preserves the spatial structure of the temperature field and correctly identifies regions of higher and lower temperature. The largest deviations occur along the left and right edges of the sheet, where the predicted temperature is slightly lower than the reference solution.

Table 13.1 reports the processing times required by the physics-based simulator and the surrogate model for a single example. The surrogate model achieves a speed-up of 150× compared with the physics-based simulator. This reduction in computational time enables its integration into the machine control pipeline and improves the efficiency of training the inverse model, since the loss function is evaluated using temperature distributions generated from the predicted machine parameters.

Table 13.1: Physics-based simulator and surrogate model runtime for example.

Simulator (s)	Surrogate model (s)	Speed-up
≈ 105.0	≈ 0.70	150×

Chapter 14

Inverse model

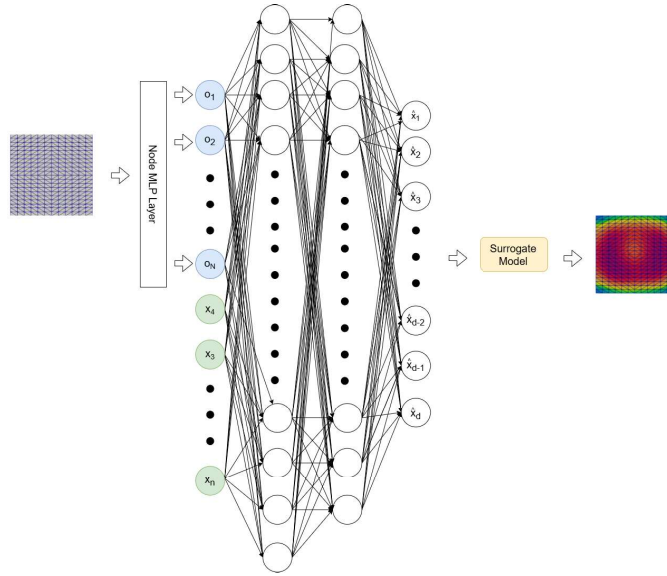


Figure 14.1: Inverse model architecture

The objective of the inverse model $\mathcal{I}$ is, given a target temperature distribution at the core of the sheet, to infer the machine parameter configuration capable of reproducing it.

In contrast to the surrogate model $\mathcal{S}$, $\mathcal{I}$ does not employ an MGNN architecture, as it is required to output continuous-valued machine parameters that are not inherently associated with mesh nodes, which instead serve as input features.

To preserve the positional temperature information associated with the mesh,

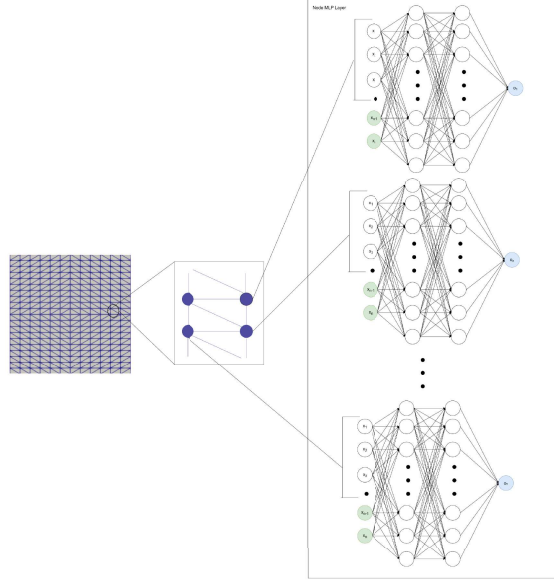


Figure 14.2: Node MLP layer

the architecture of $\mathcal{I}$ processes node-level information through a Node MLP layer, as illustrated in Figure 14.2, followed by a fully connected network, as shown in Figure 14.1.

The input size of the Node MLP layer remains fixed across the dataset. All instances share the same number of mesh nodes, while the mesh is geometrically stretched or compressed according to the sheet geometry of each example.

The Node MLP layer consists of a distinct MLP for each mesh node, each composed of two fully connected hidden layers with 16 units. The input to each node-specific MLP includes the mesh node coordinates, the desired temperature at the corresponding node, the sheet geometry (width, height, thickness), and the material properties (volumetric heat capacity, surface emissivity, and thermal conductivity). The sheet characteristics are repeated for each node, as this redundancy has been observed to yield better performance during training than excluding this information.

Each node MLP produces a single output $\mathbf{o}_i$, representing the processed information for node i of the mesh. These outputs are then concatenated with the geometric and material properties of the sheet and processed by a subsequent fully

connected network consisting of two hidden layers with 512 units each.

The inverse model predicts a 245-dimensional vector corresponding to the input required by $\mathcal{S}$, excluding the sheet geometry, and material properties, which are shared inputs between both models.

A key component of the output concerns the lamp powers configuration. In the real industrial scenario, machine operators always deactivate lamps outside the sheet area, while setting boundary lamps to the maximum capacity. This prior knowledge is embedded into the model through a masking mechanism applied during training, which prevents updates to the weight corresponding to the already known lamps values, improving convergence stability.

The model is trained using the loss function $\mathcal{L}_i$ expressed in Eq. 14.5 and optimized using Adam with a learning rate of 1×10^{-2} . Training is performed with a batch size of 64 for up to 1000 epochs, with early stopping selecting the final model at epoch 842. The training is conducted on the same hardware setup used for the surrogate model and requires approximately four days.

14.1 Loss function formulation

The inverse problem is inherently ill-posed, since multiple machine configurations may produce similar temperature distributions. Therefore, the objective is not exact reconstruction of simulator inputs, but rather the identification of physically consistent and cost-efficient solutions.

The inverse model prediction $\hat{\mathbf{x}}$ is evaluated through the surrogate model as $\hat{\mathbf{y}} = \mathcal{S}(\hat{\mathbf{x}})$. From the obtained output $\hat{\mathbf{y}}$, we extract the predicted core temperature distribution $\hat{\mathbf{y}}_C$, the maximum temperatures at the top and bottom of the sheet ($\hat{\mathbf{y}}_T, \hat{\mathbf{y}}_B$), the net heating time $\hat{\mathbf{y}}_H$, and the energy consumption $\hat{\mathbf{y}}_E$.

The temperature error is defined as:

$$E_T = \frac{1}{n} \sum_{i=1}^n \left(\hat{\mathbf{y}}_{C,i} - \left(\mathbf{y}_{C,i} + \frac{\epsilon}{2} \right) \right)^2 \quad (14.1)$$

where $\epsilon = 3$ °C defines a temperature tolerance. Shifting the target by $\epsilon/2$ introduces a safety margin that encourages the model to avoid under-heating, which

was observed during preliminary experiments.

Process cost is computed as:

$$P_{\text{Cost}} = \frac{1}{n} \sum_{i=1}^n \left(\frac{h}{3600} \hat{\mathbf{y}}_{H,i} + e \hat{\mathbf{y}}_{E,i} \right) \quad (14.2)$$

where h and e are machine-hour cost and energy cost coefficients, respectively.

To ensure material safety constraints, we introduce a penalty term for configurations in which the top surface maximum temperature exceeds the maximum core temperature by more than $\gamma = 15^\circ\text{C}$:

$$\lambda = \frac{1}{n} \sum_{i=1}^n \max((\hat{\mathbf{y}}_{T,i} - \hat{\mathbf{y}}_{C,i}^{\max} - \gamma)^2, 0) \quad (14.3)$$

Only the top surface temperature is considered in this penalty term, as preventing overheating on the top surface, which corresponds to the visible side of the final product, is of primary importance. In contrast, the bottom surface is in contact with the mould during forming and is therefore less critical in terms of thermal degradation.

As for the surrogate model, a sigmoid weighting term balances temperature distribution accuracy and cost minimization:

$$\sigma = \text{sigmoid} \left(\frac{E_T - \epsilon^2}{\tau \cdot \epsilon^2} \right) \quad (14.4)$$

The final loss is defined as:

$$\mathcal{L}_i = \sigma E_T + (1 - \sigma) P_{\text{Cost}} + \lambda \quad (14.5)$$

This formulation ensures that the model prioritizes temperature accuracy when errors are significant and progressively shifts its focus toward cost efficiency once thermal targets are sufficiently satisfied, while maintaining safety constraints throughout training.

14.2 Model assessment

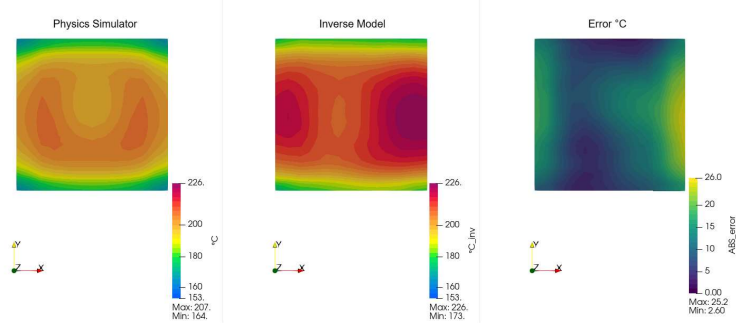


Figure 14.3: Prediction example on the test set. From left to right: the temperature distribution obtained from the physics-based simulator, the prediction produced by the inverse model, and the absolute error between the target temperature distribution and the predicted one.

At the end of training, evaluation of the temperature distribution revealed a maximum average absolute error of 15.33°C on the training set, 15.36°C on the validation set and 22.99°C on the test set.

Figure 14.3 illustrates the same test case shown in Figure 13.2, using the inverse model prediction. The model successfully captures the spatial heating strategy, increasing temperatures along the lateral regions while maintaining lower values in the central region and along the top and bottom edges of the sheet.

The error is mainly concentrated on the right-hand side of the sheet, while remaining low in the central and boundary regions. This temperature deviation arises from the inverse model’s tendency to favor higher lamp power configurations, reflecting the learned trade-off between energy usage and processing time, where shorter high-power heating cycles are more cost-efficient than prolonged low-power operation.

Quantitatively, the inverse model achieves an average cost reduction of 49.53% on the training set, of 49.40% on the validation set and of 34.52% on the test set.

Although the results are satisfactory, since the model is trained on simulator-generated data, reducing the temperature deviation with respect to the target distribution is important to improve robustness when applied to real industrial scenarios, where sheet characteristics and operating conditions may vary across

cases.

For this reason, a targeted optimization step is applied at the individual-instance level, starting from the inverse model output. This step primarily reduces the temperature distribution error and subsequently refines the process cost, balancing the trade-off between thermal accuracy and cost minimization.

Chapter 15

Optimization

The objective of the optimization procedure is to refine the inverse model prediction by first reducing the temperature distribution error and subsequently minimizing the process cost. The procedure is designed for practical deployment, allowing direct use by the machine operator through an interface exposing the main control parameters.

The optimization procedure implemented is described in Algorithm 5. The algorithm uses the Adam optimizer [25] for 1000 iterations with a learning rate of 1×10^{-1} . It takes as input the initial parameter values predicted by $\mathcal{I}$, the maximum allowable temperatures for the top and bottom surfaces of the sheet, denoted by $T_T^{\max}$ and $T_B^{\max}$, respectively, the machine-hour cost h , and the energy cost e . The output of the algorithm is the optimized set of machine parameters $\mathbf{x}^*$.

The temperature thresholds, together with the machine-hour and energy costs, can be defined by the machine operator based on the material properties, mold geometry, and the cost optimization objective, such as shorter processing times or reduced energy consumption. Since the optimization is performed at the instance level, these values can be specified more precisely for each case, allowing the optimization to better adapt to the specific problem.

The optimization algorithm is divided into two stages of 500 iterations each. The first stage focuses on reducing the temperature distribution error using Eq. 15.3 as the objective function. The second stage aims to minimize the process cost while

Algorithm 5 Inverse model process parameters optimization

Require: Initial machine parameters $\hat{\mathbf{x}} = \mathcal{I}(\hat{\mathbf{y}})$, maximum allowable temperatures $T_T^{\max}$ and $T_B^{\max}$, machine-hour cost h and energy cost e

Ensure: Optimized machine parameters $\mathbf{x}^*$

```
1:  $I_1 = 500$ 
2:  $I_2 = 1000$ 
3: Initialize the Adam optimizer
4: Set  $\mathbf{x}^{(0)} \leftarrow \hat{\mathbf{x}}$ 
5: for  $k = 1$  to  $I_1$  do ▷ First stage
6:    $\hat{\mathbf{y}} \leftarrow \mathcal{S}(\mathbf{x}^{(k-1)})$ 
7:   Extract  $\hat{\mathbf{y}}_C$ ,  $\hat{\mathbf{y}}_T$ , and  $\hat{\mathbf{y}}_B$ 
8:   Compute  $E_T^{\max}$  according to Eq. 15.1
9:   Compute  $\lambda_T$  and  $\lambda_B$  according to Eq. 15.2
10:  Evaluate objective function  $f_{\text{first}}$  according to Eq. 15.3
11:  Update of  $\mathbf{x}^{(k)}$  by minimizing  $f_{\text{first}}$  with Adam
12: end for
13: for  $k = I_1$  to  $I_2$  do ▷ Second stage
14:    $\hat{\mathbf{y}} \leftarrow \mathcal{S}(\mathbf{x}^{(k-1)})$ 
15:   Extract  $\hat{\mathbf{y}}_C$ ,  $\hat{\mathbf{y}}_T$ , and  $\hat{\mathbf{y}}_B$ 
16:   Compute  $E_T^{\max}$  according to Eq. 15.1
17:   Compute  $\lambda_T$  and  $\lambda_B$  according to Eq. 15.2
18:   Evaluate objective function  $f_{\text{first}}$  according to Eq. 15.3
19:   Compute  $\sigma$  according to Eq. 15.4
20:   Evaluate objective function  $f_{\text{second}}$  according to Eq. 15.5
21:   Update of  $\mathbf{x}^{(k)}$  by minimizing  $f_{\text{second}}$  with Adam
22: end for
23:  $\mathbf{x}^* \leftarrow \mathbf{x}^{(1000)}$ 
24: return  $\mathbf{x}^*$ 
```

preserving the temperature accuracy achieved in the first stage, using Eq. 15.5 as the objective function.

In the first-stage of the optimization, during the initialization in lines 1–4 of Algorithm 5, the Adam optimizer is initialized, and the machine parameters predicted by the inverse model $\hat{\mathbf{x}}$, are set as the starting point of the optimization.

As illustrated in lines 5–12 of Algorithm 5, the surrogate model $\mathcal{S}$ is evaluated for the current machine parameters, and the resulting output prediction are used to compute the temperature error.

The temperature error considered is computed as the maximum absolute deviation between the predicted and target temperature distributions, incorporating a tolerance margin $\epsilon = 3$ °C, consistently with the error formulation used in the inverse model (see Section 14.1):

$$E_T^{\max} = \max \left| \hat{\mathbf{y}}_C - \left(\mathbf{y}_C + \frac{\epsilon}{2} \right) \right| \quad (15.1)$$

overheating is penalized when the predicted top and bottom temperatures exceed the allowable limits:

$$\lambda_T = \max(\hat{\mathbf{y}}_T - T_T^{\max}, 0), \quad \lambda_B = \max(\hat{\mathbf{y}}_B - T_B^{\max}, 0) \quad (15.2)$$

The first-stage optimization objective function is defined as:

$$f_{\text{first}} = E_T^{\max} + \lambda_T + \lambda_B. \quad (15.3)$$

During the second stage of the optimization, as reported in lines 13–22 of Algorithm 5, the output obtained from the first-stage is used as the initial input. In this case the objective is to reduce the process cost while preserving the temperature accuracy achieved at the end of the first stage. To this end, a sigmoid function is used to modulate the trade-off between temperature accuracy and process cost minimization:

$$\sigma = \text{sigmoid} \left(\frac{E_T^{\max} - \epsilon}{\tau \cdot \epsilon} + \lambda_T + \lambda_B \right) \quad (15.4)$$

The objective function minimized during the second stage is:

$$f_{\text{second}} = \sigma f_{\text{first}} - (1 - \sigma) \cdot \frac{1}{P_{\text{Cost}}} - \epsilon. \quad (15.5)$$

where P_{Cost} is the cost function defined in Eq. 14.2.

Finally, after completion of the two optimization stages, the resulting machine parameters are returned, as reported in lines 23–24 of Algorithm 5.

15.1 Evaluation

To evaluate the effectiveness of the inverse model initialization, we compare the maximum absolute temperature error obtained after optimization when initialized with the parameters predicted by $\mathcal{I}$ against a random initialization. The analysis

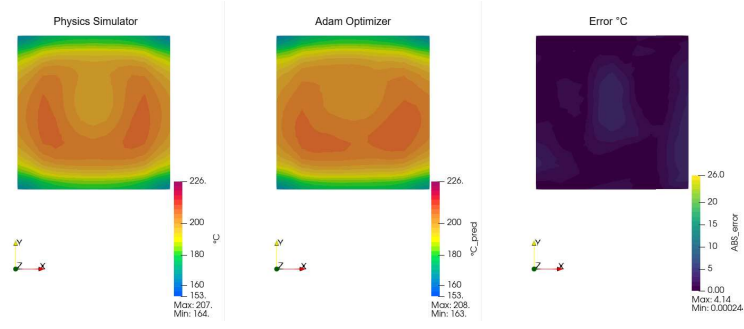


Figure 15.1: Prediction example on the test set. From left to right: the temperature distribution obtained from the physics-based simulator, the prediction produced by the optimizer, and the absolute error between the target temperature distribution and the predicted one.

is conducted using paired comparisons on both validation and test sets, in order to assess the behavior of the optimizer on both simulated and real data.

The results were obtained by executing the optimization strategy, taking into account the material-specific thresholds indicated in [57] when selecting the maximum allowable temperatures $T_T^{\max}$ and $T_B^{\max}$. To allow for comparison with the results obtained from the inverse model, the values of h and e in Eq. 14.2 were set to the same values used by the inverse model and kept constant across all examples.

On the validation set, initialization from the inverse model yields a lower maximum absolute error than random initialization in 80% of the cases, with a statistical significance of $p_{value} = 9.132e - 54$, computed using the *Wilcoxon signed-rank test* [79]. On the test set, this proportion is 67%, with $p_{value} = 3.015e - 5$.

These results indicate that initialization from the inverse model provides a more favorable starting point for the optimization procedure than random initialization. Specifically, after the optimization procedure starting from the inverse model outputs, we achieved a maximum average absolute error of 3.15°C on the validation set and 4.23°C on the test set.

Figure 15.1 illustrates the same example shown in Figure 13.2 and Figure 14.3, considering the prediction obtained after optimization. The error previously observed along the right-hand side of the sheet is significantly reduced, while the remaining deviations are mainly concentrated in the central region, where the

predicted temperature is slightly higher than the target.

The optimizer maintains the tendency to favor higher heating power over shorter durations, reflecting the trade-off learned by the inverse model between energy consumption and processing time. However, compared to the inverse model, the optimization step places greater emphasis on reducing the temperature distribution error by allowing longer heating times. As a result, the additional cost reduction is smaller than that achieved by the inverse model, with a reduction of 11.77% on the validation set and 8.98% on the test set.

Chapter 16

Conclusion

This second part of the thesis presented a data-driven methodology for optimizing the heating phase in thermoforming, considering an industrial thermoforming machine and addressing the inverse problem of determining the machine parameters required to produce a desired core temperature distribution.

The proposed approach relies on two neural network models. A surrogate model based on a Mesh Graph Neural Network (MGNN) is first developed to approximate a physics-based simulator, enabling fast and accurate prediction of the temperature distribution at the core of the sheet. The model achieves a speed-up of approximately $150\times$ with respect to the simulator, while maintaining a maximum average absolute error of $3.43\text{ }^{\circ}\text{C}$ on the validation set and $9.65\text{ }^{\circ}\text{C}$ on the test set.

Building on this model, an inverse model is introduced to predict machine parameters from a target temperature distribution, while incorporating process cost considerations. The model significantly reduces process cost and processing time by favoring higher lamp power configurations that allow the desired thermal state to be reached more quickly. This strategy produces temperature distributions that qualitatively match the desired spatial patterns, but leads to higher predicted temperatures compared to the target distribution. Consequently, the maximum average absolute errors of $15.36\text{ }^{\circ}\text{C}$ on the validation set and $22.99\text{ }^{\circ}\text{C}$ on the test set reflect this trade-off between heating intensity and process duration.

Since the model is trained on simulator-generated data, reducing these tem-

perature deviations is important for improving its robustness when applied to real industrial scenarios, where operating conditions and sheet characteristics may vary significantly between cases.

To this end, a two-stage optimization procedure based on the Adam optimizer is applied to refine the inverse model predictions at the individual sample level. The first stage focuses on reducing the temperature distribution error, while the second improves process cost without degrading thermal accuracy. After optimization, the maximum average absolute error is reduced to 3.15 °C on the validation set and 4.23 °C on the test set, demonstrating an improvement in temperature accuracy that, however, comes at the expense of reduced cost savings compared with the inverse model.

The results highlight the complementary roles of the two components: the inverse model provides satisfactory temperature distributions and cost-efficient initial solutions, while the optimization step refines them by reducing temperature deviations, thereby improving generalization to varying conditions.

The industrial partner has decided to start integrating the proposed strategy into thermoforming machine control workflows and is investigating its potential for extension to support other stages of the thermoforming process.
